

Cytosine Methylation Enhances DNA Condensation Revealed by Equilibrium Measurements Using Magnetic Tweezers

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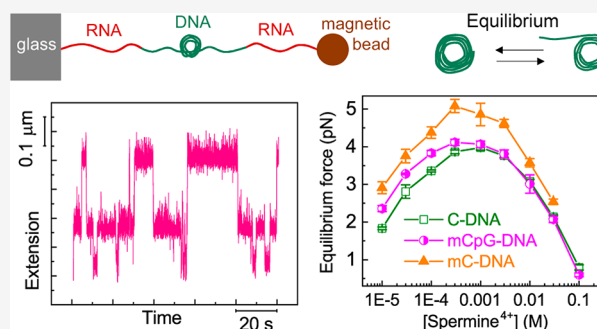


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Supporting Information

ABSTRACT: CpG methylation of DNA is common in mammalian cells. In sperm, the DNA has the highest level of CpG methylation and is condensed into toroidal structures. How CpG methylation affects DNA structures and interactions is important to understand its biological roles but is largely unknown. Using an RNA–DNA–RNA structure, we observed the equilibrium hopping dynamics between the condensed and extended states of DNA in the presence of polyamines or polylysine peptide as a reduced model of histone tails. Combining with the measured DNA elasticities, we report that CpG methylation of each cytosine nucleotide substantially increases DNA–DNA attraction by up to $0.2 k_B T$. For the DNA with 57% GC content, the relative increase caused by CpG methylation is up to 32% for the spermine-induced DNA–DNA attraction and up to 9% for the polylysine-induced DNA–DNA attraction. These findings help us to evaluate the energetic contributions of CpG methylation in sperm development and chromatin regulation.



■ INTRODUCTION

Despite strong repulsion between negatively charged backbones, DNA can be condensed by many cationic counterions, such as high-valent ($N \geq 3$) cations, cationic surfactants, lysine/arginine-rich peptides, and cationic proteins.^{1,2} DNA condensation and decondensation are properly regulated in cells to facilitate DNA packing, DNA replication, and gene transcription. How to effectively control DNA condensation is very important in many biological processes and can be useful in applications, such as gene delivery. DNA condensation should depend on not only the external conditions of DNA molecules but also the chemical modification to DNA. Methylation is arguably the most important chemical modification to DNA. How methylation affects DNA condensation, however, remains largely unclear.

In mammals, most DNA methylation occurs at cytosine nucleotides at CpG (a cytosine nucleotide followed by a guanine nucleotide) sites. The CpG methylation is tied strongly to cell development, pluripotency, carcinogenesis, transcriptional regulation, chromatin remodeling, aging, cancer, etc.^{3–6} The level of CpG methylation is the highest in sperm where DNA is condensed into close-packed toroidal subunits.⁴ In somatic cells, the DNA has well-regulated CpG methylation and is packed into nucleosomes, then further packed into chromosomes. Thus, it is of great interest to know whether methylation plays a role in DNA condensation.

Single-molecule techniques have been demonstrated to be effective tools to study the mechanism and kinetics of DNA

condensation as well as the elasticities of DNA in the presence of condensing agents.^{7–18} Recently, single-molecule fluorescence experiments and molecular dynamics simulations have revealed that the methyl groups could relocate the high-valent cations from major grooves of DNA to interhelical regions, thereby increasing DNA–DNA attraction.^{19,20} In this work, to quantify the effects of CpG methylation, we will study the condensation of methylated DNA at equilibrium in the presence of high-valent cations using single-molecule magnetic tweezers. Combining with the measured elasticities of DNA before condensation, we will calculate the change in DNA–DNA attraction that drives DNA condensation caused by CpG methylation. These results will help us to evaluate how CpG methylation affects the development of sperm and the regulation of chromatin dynamics.

■ MATERIALS AND METHODS

We tethered a single DNA between an antidigoxigenin coated cover-glass and a streptavidin-coated superparamagnetic bead (Figure 1a). After bead tethering, we passivated the glass and bead surfaces using 10% w/v MPEG-Succinimidyl Valerate ester (MW 2K, Hunan

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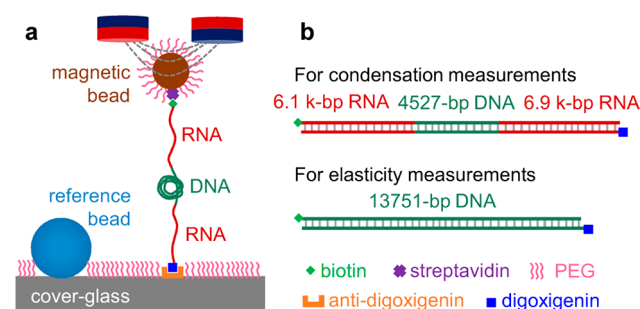


Figure 1. Experimental setup. (a) Home-built magnetic tweezers. (b) DNA constructs for condensation and elasticity measurements.

Huateng Pharmaceutical) for 2 h. We acquired all data at 10 mM Tris-HCl pH 7.5 and 22 °C. See more details of the home-built magnetic tweezers in our previous works.^{21–27}

Taking advantage of double-stranded (ds) RNA that resists condensation,²⁸ we used two long dsRNA handles to protect the inside 4527 bp dsDNA from possible interactions with the substrate surfaces in DNA condensation experiments (Figure 1b). In DNA elasticity measurements, we used a 13751 bp long dsDNA to achieve high precision. The nonmethylated DNA (C-DNA) was amplified through ordinary PCR using KOD-plus DNA polymerase (TOYOBO). We added methyl groups to the cytosine nucleotides at CpG sites (mCpG-DNA) using CpG Methyltransferase (New England Biolabs). To magnify the effects of cytosine methylation for detection, we added methyl groups to all cytosine nucleotides (mC-DNA) through PCR using 5-methyl-dCTP (New England Biolabs). Please find the protocols to prepare the DNA constructs in the Supporting Information (Figure S1a,b) and our previous work.²³ The 4527 bp DNA contains 2575 cytosine nucleotides, out of which 828 are located at CpG sites. The 13751 bp DNA contains 7865 cytosine nucleotides, out of which 1166 are located at CpG sites. The efficiency of cytosine methylation was confirmed using BstUI (New England Biolabs) whose cleavage ability was blocked by CpG methylation (Figure S1c,d).

We purchased Hexamine cobalt(III) chloride (CoHex³⁺), Spermidine-3HCl (Spermidine³⁺), and Spermine-4HCl (Spermine⁴⁺) from Sigma-Aldrich. These high-valent cations were widely used to study DNA condensation and DNA–DNA interactions.^{13–20,29–32} We synthesized the polylysine peptide Hexa-lysine-6HCl (PLL⁶⁺) in Sangon Biotech as a reduced model of the highly charged, intrinsically disordered, lysine-rich histone tails.²⁰

To determine the nucleation (F_n) and equilibrium (F_e) condensation forces, we stretched the RNA–DNA–RNA molecule by repeating force-cycles in the presence of high-valent cations. Each cycle contained four force regions (Figure 2a): (i) A force-decreasing region where the applied force was successively kept at a series of constant forces reduced at -0.01 pN/s and during force-decreasing the DNA nucleated and condensed; (ii) A region where a low constant force was kept for 20 s; (iii) A force-increasing region where the applied force was successively kept at a series of constant forces raised at 0.01 pN/s and during force-increasing the DNA passed by the equilibrium point and then completely decondensed at even high forces; (iv) A region where a high constant force was kept for 10 s. In the force-decreasing region, the reduction in extension during condensation was always slightly shorter than the extension of 4527 bp DNA, indicating the RNA handles did not condense. In the force-increasing region and force-decreasing region, we calculated the average and variance of the extension at each constant force (Figure 2b). We determined F_n and F_e by an abrupt increase in the variance of molecule extension.^{24–26}

RESULTS

The Kinetics of DNA Condensation under Tension. Whether methylated or not, the DNA condensed in a one-step manner in the force-decreasing region (Figure 3), which can be

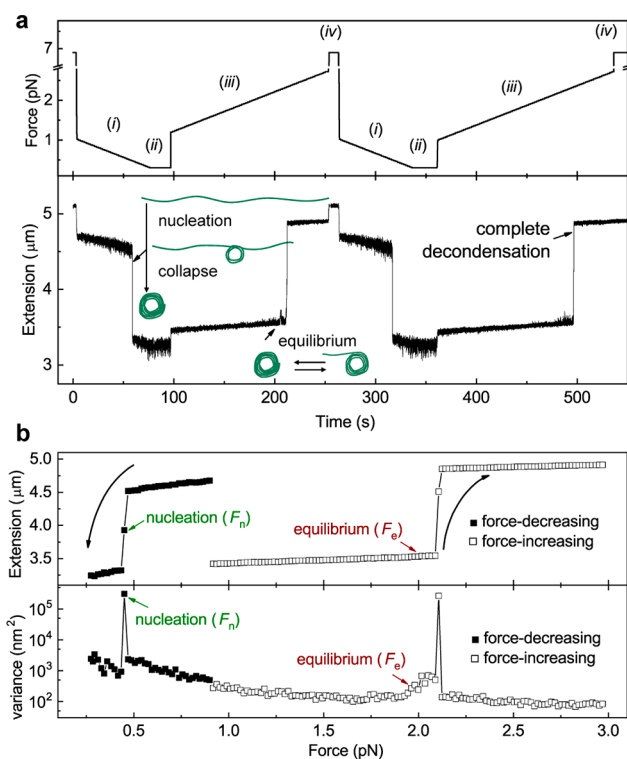


Figure 2. Determining the condensation forces by repeating force-cycles. The representative curves were obtained at 0.2 mM CoHex³⁺ using C-DNA. (a) The time-extension course recorded during repeating force-cycles. (b) The nucleation (F_n) and equilibrium (F_e) condensation forces marked by the abrupt increase in the variance of molecule extension.

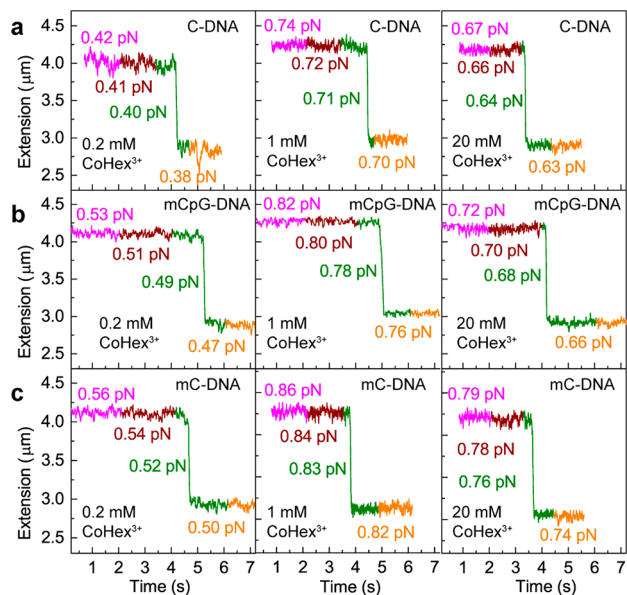


Figure 3. One-step kinetics from the extended state to the condensed state of DNA during force-decreasing.

explained by the nucleation-collapse mechanism.^{15,16} To overcome the large energy barriers, the nucleation occurred at forces much lower than the energy equilibrium. Immediately after nucleation, the DNA rapidly condensed against the low nucleation force.

After DNA condensation, we stretched the condensed DNA through a series of increasing constant forces near the equilibrium (Figure 2). To detect the kinetics more clearly, we held each constant force for hundreds of seconds (Figure 4).

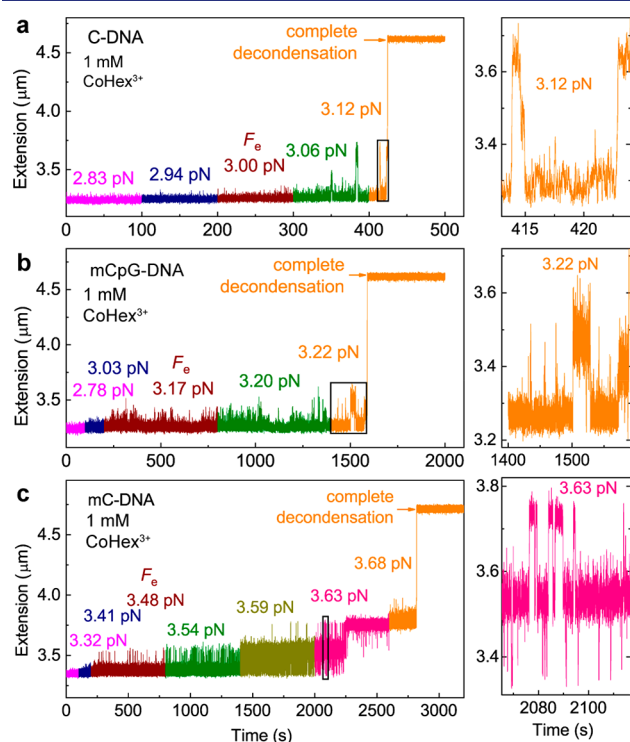


Figure 4. The hopping kinetics between the extended and condensed states of DNA at constant forces. The detailed time-courses in the black rectangles were magnified for clarity.

For all of the C-DNA, mCpG-DNA, and mC-DNA, we observed the hopping kinetics of DNA extension near the equilibrium at constant forces (magnified panels), which indicates the coexistence of the extended and condensed state of DNA. Forces of about 0.2 pN higher than F_e (wine) led to complete decondensation of the DNA.

Cytosine Methylation Enhances DNA Condensation.

With the increase in the concentration of CoHex³⁺, Spermidine³⁺, or Spermine⁴⁺, both F_n and F_e first increase and then decrease (Figure 5). The reverse of trend is probably due to the charge-inversion of DNA caused by the excessive binding of high-valent cations.^{15,16} The peak values of F_n and F_e increase with the valence of cations (PLL⁶⁺ > Spermine⁴⁺ > CoHex³⁺ > Spermidine³⁺). We only measured until 0.1 mM PLL⁶⁺ as dsRNA handles might condense or interact with the glass/bead surfaces at higher PLL⁶⁺ concentrations (Figure S2). Such RNA condensation or RNA-surface interaction may disturb the measurement of DNA condensation.

Figure 5 compares DNA condensation for DNA samples with three levels of cytosine methylation: C-DNA, mCpG-DNA, and mC-DNA. For all the four cations, cytosine methylation enhances DNA condensation as indicated by the increases in F_n and F_e . In Figure 5, we used the loading/unloading rates of ± 0.01 pN/s. To determine the effects of loading/unloading rates, we further measured F_n and F_e at ± 0.001 and ± 0.003 pN/s in the presence of 5 mM CoHex³⁺ (Figure S3). We found F_n slightly increased at low rates whereas F_e kept constant, which were consistent with the

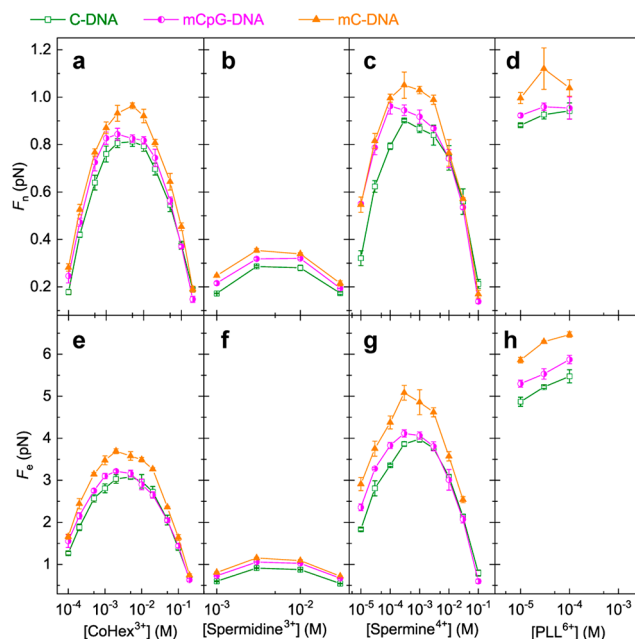


Figure 5. Effects of cytosine methylation on the nucleation and equilibrium condensation forces. At each cation condition, the averages and standard errors of F_n and F_e obtained from more than three molecules were plotted as data points and error bars. We only measured until 0.1 mM PLL⁶⁺ to avoid RNA condensation.

nonequilibrium nature of F_n and the equilibrium nature of F_e . The rate-independent F_e will be used to quantify the enhancement of DNA–DNA attraction caused by cytosine methylation.

DNA Elasticities in the Presence of High-Valent Cations. To obtain the changes in DNA extension during DNA condensation, we next determined DNA elasticities in the presence of high-valent cations using the 13751 bp DNA. To avoid DNA condensation and DNA–surface interactions, we only measured the force–extension curves of DNA at higher forces (Figure S4). We fitted each force–extension curve to the extensible worm-like chain model:³³

$$\frac{x}{NL} = 1 - \sqrt{\frac{k_B T}{4FP}} + \frac{F}{K} \quad (1)$$

Here, N is the number of base pairs of the DNA. The measured variables x is the DNA extension and F is the applied force. The fitting parameters L is the contour length per base pair, P is the bending persistence length, and K is the stretching modulus.

Figure 6 shows that in the cases of CoHex³⁺, Spermidine³⁺, and Spermine⁴⁺, P , L , and K decrease first and then increase with the increase in cation concentrations. The reverse of the trend may be attributed to charge-inversion of DNA caused by excessive binding of high-valent cations.^{16,27} In the case of PLL⁶⁺, both P and L decrease and then reach plateaus, while K keeps increasing. We did not further increase the concentration of PLL⁶⁺ to avoid DNA–surface interactions.

The enhancement of the bending flexibility of an isolated DNA will be critical especially in nanoscale bending of DNA fragments, a prerequisite process of DNA condensation.^{1,12} The shortening of DNA contour length means the same spatial length of DNA contains more DNA base pairs (i.e., more charge interactions between DNA base pairs). Thus, high-valent cations soften and shorten DNA, and both of the effects

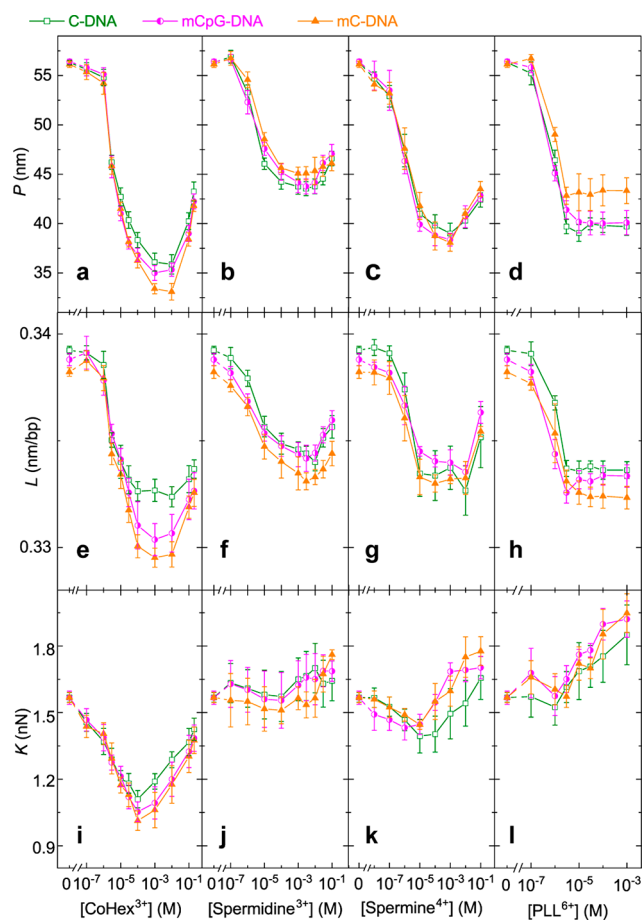


Figure 6. The elasticities of DNA in the presence of high-valent cations. At each cation condition, the average and standard error of bending persistence length (P), contour length per base pair (L), and stretching modulus (K) obtained from more than three molecules were plotted as data points and their error bars.

benefit DNA condensation. Figure 6 shows that cytosine methylation slightly softens and shortens DNA in the presence of CoHex^{3+} , but not in the presence of Spermine^{4+} , Spermidine^{3+} , or PLL^{6+} .

Quantifying the Enhancement of DNA–DNA Attraction Caused by Cytosine Methylation. On the basis of the measured equilibrium DNA condensation force (Figure 5e–h) and DNA elasticities (Figure 6), we calculated the DNA–DNA attraction that drives DNA condensation. At the equilibrium, the extended and condensed DNA states coexist, which gives

$$\Delta G = \Delta G_{\text{chem}} - l \cdot F_e = 0 \quad (2)$$

Here, ΔG is the total free energy difference per base pair between the extended and condensed states of DNA at the equilibrium; ΔG_{chem} is the DNA–DNA attraction without external force, which includes contributions of electrostatic interactions and bending of DNA; F_e is the equilibrium condensation force (Figure 5e–h); l is the DNA extension per base at F_e , which depends on the elasticity of DNA for the given condition (i.e., methylation status and cation concentration). We calculated l using the extensible worm-like chain model with determined values of P , L , and K (Figure 6).

As shown in Figure 7a–d, we calculated the ΔG_{chem} for C-DNA, mCpG-DNA, and mC-DNA. To determine the effects of cytosine methylation, we further calculated the changes in

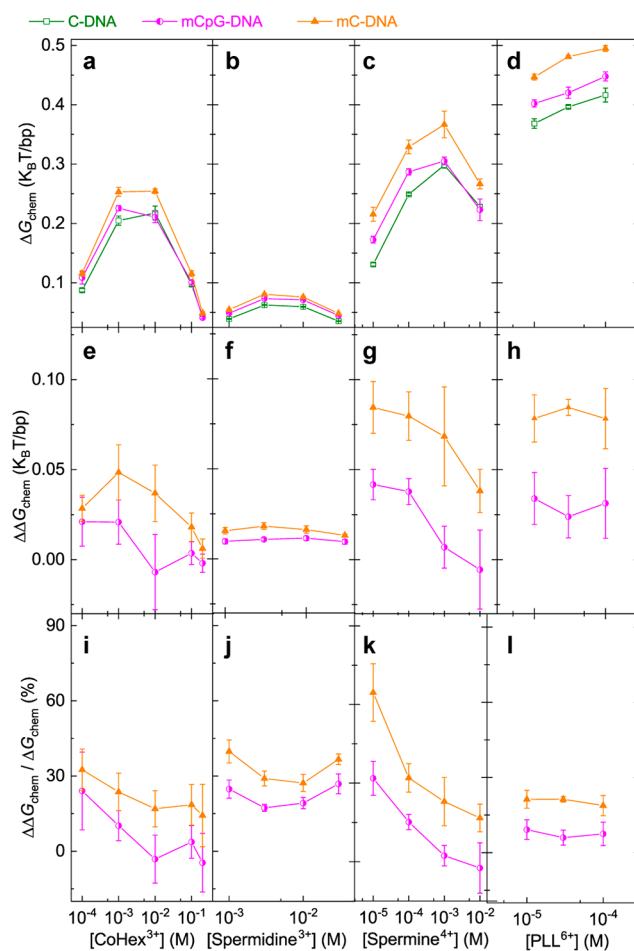


Figure 7. Cytosine methylation increases DNA–DNA attraction. At each cation condition, the average and standard error of DNA–DNA attraction calculated from more than three molecules were plotted as a data point and its error bar. (a–d) The DNA–DNA attraction per base pair without external force (ΔG_{chem}) for C-DNA, mCpG-DNA, and mC-DNA. (e–h) The change in ΔG_{chem} caused by cytosine methylation. (i–l) The percentage change in ΔG_{chem} caused by cytosine methylation.

ΔG_{chem} caused by cytosine methylation ($\Delta \Delta G_{\text{chem}}$, Figure 7e–h). We found $\Delta \Delta G_{\text{chem}}$ was larger at lower concentrations of high-valent cations. At 0.01 mM Spermine^{4+} , $\Delta \Delta G_{\text{chem}}$ approaches $\sim 0.04 k_B T/\text{bp}$ for mCpG-DNA ($\sim 0.23 k_B T$ per methylated cytosine nucleotide) and $\sim 0.08 k_B T/\text{bp}$ for mC-DNA ($\sim 0.15 k_B T$ per methylated cytosine nucleotide). At 0.01 mM PLL^{6+} , $\Delta \Delta G_{\text{chem}}$ approaches $\sim 0.03 k_B T/\text{bp}$ for mCpG-DNA ($\sim 0.18 k_B T$ per methylated cytosine nucleotide) and $\sim 0.08 k_B T/\text{bp}$ for mC-DNA ($\sim 0.14 k_B T$ per methylated cytosine nucleotide). It appears that cytosine methylation at CpG sites increases DNA–DNA attraction more significantly than cytosine methylation at other sites. As shown in Figure 7i–l, at 0.01 mM Spermine^{4+} , 5mCpG increases ΔG_{chem} by $\sim 32\%$ and 5mC further increases ΔG_{chem} by $\sim 65\%$ for our $\sim 57\%$ GC DNA fragment. At 0.01 mM PLL^{6+} , 5mCpG increases ΔG_{chem} by $\sim 9\%$ and 5mC further increases ΔG_{chem} by $\sim 21\%$.

DISCUSSION

In previous single-molecule DNA condensation experiments, the DNA was directly tethered between the bead and glass surfaces and the decondensation showed step-wise kinetics in a

wide range of applied force. The complete decondensation cannot be accomplished even at the force of 23 pN in the presence of CoHex³⁺.¹⁸ By using long dsRNA handles to protect the inside dsDNA from DNA–surface interactions, we observed the hopping kinetics between the condensed and extended states of DNA at constant forces at the equilibrium (Figure 4). Forces of ~ 0.2 pN higher than the equilibrium force lead to complete decondensation of the DNA (Figure 4). The equilibrium condensation force depends on the concentration of high-valent cations and achieves a maximum of ~ 3.2 pN for CoHex³⁺, ~ 0.9 pN for Spermidine³⁺, and ~ 4.0 pN for Spermine⁴⁺ using C-DNA. Using the RNA–DNA–RNA structure, our nucleation force achieves a maximum of ~ 0.8 pN for CoHex³⁺, ~ 0.3 pN for Spermidine³⁺, and ~ 0.9 pN for Spermine⁴⁺ using C-DNA, which are far below the previous values obtained from the directly tethered DNA molecules for the same high-valent cations.^{13,15–18} Thus, if DNA was directly tethered to the surfaces, the DNA–surface interactions might disturb the DNA condensation experiments.

Biological regulations often operate at margin zones of phase transition. The enhancement of DNA–DNA attraction caused by CpG methylation may make a deterministic contribution to DNA condensation under certain conditions, especially when the concentrations of high-valent cations are close to the threshold for DNA condensation. Using $\Delta\Delta G_{\text{chem}}$ of $0.03 k_{\text{B}}T/\text{bp}$ and considering that the perimeter of the toroids is 100–300 nm (about 300–1000 bp), 5mCpG increases the DNA–DNA attraction by about 9–30 $k_{\text{B}}T$ for each circle in the toroids. This increase in DNA–DNA attraction caused by 5mCpG is strong enough to suppress the thermal energy, which should make a valid contribution to DNA condensation. The enhancement of G_{chem} by CpG methylation may explain why sperm DNA has the highest level of CpG methylation from the thermodynamic perspective. Recall that PLL⁶⁺ serves as a reduced model of histone tails that modulate nucleosome organizations and interactions. In the presence of PLL⁶⁺, methylation of each cytosine nucleotide at CpG sites increases ΔG_{chem} by up to $\sim 0.18 k_{\text{B}}T$, which may be related to the role of 5mCpG in the subtle regulation of nucleosome stability revealed in previous experiments.^{34–39}

It is reasonable that cytosine methylation increases the DNA–DNA attraction in the presence of high-valent cations. As revealed in the previous simulations, cytosine methylation provides the exclusion-volume effects in grooves for suppressing internal cation-binding deeply in grooves and correspondingly increases the external cation-binding around phosphate groups.¹⁹ Such stronger external cation-binding would favor the high-valent cation-bridging effect between DNA chains and increase DNA–DNA attraction.^{19,40–42} As cytosine methylation mostly had no detectable effects on the elasticity of DNA in the presence of high-valent cations (Figure 6), likely the enhancement in G_{chem} are dominantly caused by changes in electrostatic interactions.

In addition to DNA condensation, our measurements may also help to evaluate the effects of cytosine methylation on other processes of DNA, including bundling, crystallization, looping, supercoiling, bridging, cross-linking, crossing over, juxtaposition, knotting, and entanglement, where direct DNA–DNA contacts are involved. DNA–DNA contacts are involved in many fundamental biological processes such as homologous recombination,⁴³ DNA packaging and ejection of virals,³² chromatin dynamics and remodeling,⁴⁴ formation of heterochromatin at highly CpG-methylated DNA,⁴⁵ trans-silencing

increased with DNA methylation,⁴⁶ promoter-enhancer interactions for transcription regulation.⁴⁷ DNA–DNA contacts are also involved in many applications such as DNA origami/nanostructures,⁴⁸ gene delivery and gene therapy,⁴⁹ DNA-guided organic semiconductor,⁵⁰ DNA aptamers and sensors.⁵¹ For example, since cytosine methylation enhances DNA condensation, the cytosine-methylated DNA might be a better “genetic cargo” in cationic liposome- or polymer-based gene delivery systems.

Recent simulations revealed that external binding of high-valent cations around phosphate groups softened DNA while the internal binding of high-valent cations deeply in grooves could counteract the above affect.^{27,52} It is understandable that cytosine methylation slightly softens DNA in the presence of CoHex³⁺ but has no detectable effect on the DNA stiffness in the presence of Spermine⁴⁺ (Figure 6a,c). The increase of external cation-binding and the decreased internal cation-binding caused by the exclusion effect of cytosine methylation would slightly soften the DNA in the presence of CoHex³⁺. As indicated in recent simulations, compared with CoHex³⁺, Spermine⁴⁺ has a slightly stronger tendency for external binding around phosphate groups and slightly less tendency for internal binding in grooves.²⁹ Thus, the exclusion effect in grooves from cytosine methylation would be slightly stronger for the binding of CoHex³⁺ than for that of Spermine⁴⁺, and resultantly, cytosine methylation can slightly soften the DNA in the presence of CoHex³⁺ while not in the presence of Spermine⁴⁺.

Controversial results about the effects of cytosine methylation on the stiffness of DNA were obtained in experiments and simulations previously. Single-molecule cyclization experiments revealed that cytosine methylation stiffened DNA.^{37–39} Nevertheless, optical/magnetic tweezing and AFM scanning experiments revealed that cytosine methylation softened DNA.^{53,54} Previous simulations reported varied results also: cytosine methylation either stiffened DNA^{37–39,55,56} or had no effect⁵⁷ or softened DNA.^{58,59} A part of the contradicting results is probably due to the concept of flexibility, which in reality depends on the nature of the distortion. Another part is probably related to the technique used and the type of observable collected. In this work, we found cytosine methylation slightly softened DNA in the presence of CoHex³⁺, while it mostly had no detectable effects on the stiffness of DNA. Thus, the effects of cytosine methylation on DNA stiffness might be sensitive to conditions such as cations, GC content, DNA length, and local sequence pattern, which calls for future systematic studies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.9b11957>.

Figures S1–S4 (PDF)

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Notes

The authors declare no competing financial interest.

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